whose elementary events happen in spacetime. The world is strange, but simple (figure 4.6).

### *Quanta 1: Information is finite*

The time has come to attempt some conclusions about what it is, precisely, that quantum mechanics tells us about the world. It isn't an easy task, because quantum mechanics is not conceptually clear and its true meaning remains controversial; but it's a necessary exercise, to gain clarity and go forward. I think that quantum mechanics has revealed three aspects of the nature of things: granularity, indeterminacy and the relational structure of the world. Let's look at each of these more closely.

The first is the existence of a fundamental *granularity* in nature. The granularity of matter and light is at the heart of quantum theory. It isn't the same granularity intuited by Democritus, however. For Democritus, atoms were like little pebbles, whereas in quantum mechanics particles vanish and reappear. But the root of the idea of the substantive granularity of the world is still to be found in ancient atomism, and quantum mechanics – strengthened by centuries of experiments, by powerful mathematics, and by its extraordinary capacity for making correct predictions – is a genuine recognition of the profound insights on the nature of things reached by the great philosopher of Abdera.

Say we make measurements on a physical system and find that the system is in a particular state. For instance, we measure the amplitude of the oscillations of a pendulum and find that it has a certain value – say, somewhere between five centimetres and six centimetres (no measurement is exact in physics). Before quantum mechanics we would have said that, since there are an infinite number of possible values between five and six centimetres (for instance 5.1 or 5.101 or 5.101001 …), then there are *infinite* possible states of motion in which the pendulum could find itself: the amount of our ignorance about the pendulum state is still infinite.

Instead, quantum mechanics tells us that between five and six centimetres there is a *finite* number of possible values of the amplitude, hence our missing information about the pendulum is *finite*.

This goes for everything in general.[fn26] Therefore, the first meaning of quantum mechanics is the existence of a limit to the *information* that can exist within a system: a limit to the number of distinguishable states in which a system can be. This limitation upon infinity – this granularity of nature glimpsed by Democritus – is the first central aspect of the theory. Planck's constant *h* measures the elementary scale of this granularity.